

MIERCUREA CIUC, Romania

WMO: 151700

Lat: 46.3714N

Lon: 25.7725E

Elev: 662

StdP: 93.62

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.6	-19.4	-24.2	0.5	-22.4	-20.9	0.6	-18.7	8.6	0.7	7.7	-0.2	0.6	330	0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	29.6	18.5	27.7	18.1	26.0	17.5	20.0	26.4	19.1	25.5	18.2	24.5	2.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	13.9	22.5	16.9	13.0	21.5	16.0	12.3	20.4	60.3	26.4	57.0	25.6	54.3	24.7	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.4	5.5	DB	-26.7	32.0	3.8	1.7	-29.4	33.2	-31.7	34.2	-33.8	35.1	-36.5	36.3	(4)
(5)			WB	-26.8	21.4	3.8	1.2	-29.6	22.3	-31.8	22.9	-33.9	23.6	-36.6	24.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	-6.1	-4.0	1.1	7.3	12.1	15.8	17.2	17.1	12.4	7.1	1.4	-4.5	(6)	
	DBStd	9.33	6.16	6.09	4.85	3.39	3.24	2.78	2.60	2.72	3.32	3.80	4.54	5.63	(7)	
	HDD10.0	2103	498	391	277	91	17	1	0	0	15	104	259	450	(8)	
	HDD18.3	4373	756	624	534	330	192	83	52	57	179	349	508	708	(9)	
	CDD10.0	815	0	0	1	11	83	174	224	219	87	14	1	0	(10)	
	CDD18.3	43	0	0	0	0	0	7	18	17	1	0	0	0	(11)	
	CDH23.3	1149	0	0	0	7	47	194	384	421	92	3	0	0	(12)	
	CDH26.7	286	0	0	0	1	3	34	107	121	19	1	0	0	(13)	
(14) Wind	WSAvg	1.6	1.4	1.5	2.0	2.0	1.9	1.8	1.9	1.6	1.7	1.4	1.4	1.3	(14)	
(15) Precipitation	PrecAvg	566	29	26	27	43	71	91	83	66	40	33	28	29	(15)	
	PrecMax	764	79	83	84	74	152	216	170	131	125	99	85	61	(16)	
	PrecMin	417	1	0	1	17	30	19	13	29	0	4	3	2	(17)	
	PrecStd	85	19	19	18	17	29	41	35	30	28	23	17	16	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	11.5	18.2	24.1	26.7	29.9	31.9	32.3	29.2	22.7	16.5	9.6	(19)	
		MCWB	5.1	6.6	9.6	13.2	15.4	18.6	18.2	19.1	16.6	14.4	11.1	7.0	(20)	
	2%	DB	4.9	7.8	14.1	20.6	24.6	27.4	29.5	29.7	25.9	20.0	13.3	6.0	(21)	
		MCWB	3.2	4.8	7.7	11.8	15.0	18.2	18.5	18.8	16.2	13.1	9.7	4.6	(22)	
	5%	DB	3.1	5.6	11.3	17.7	22.6	25.6	27.4	27.6	23.5	17.2	10.7	4.0	(23)	
		MCWB	1.9	3.4	6.2	10.0	14.4	17.4	18.4	18.2	15.2	12.1	8.0	2.9	(24)	
	10%	DB	1.5	3.6	8.6	15.5	20.7	23.7	25.3	25.6	21.0	14.7	8.3	2.1	(25)	
		MCWB	0.6	2.1	5.0	9.2	13.7	16.7	17.7	17.7	14.4	10.8	6.5	1.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.5	7.2	10.2	13.8	17.2	20.7	21.1	21.4	18.4	15.4	11.7	7.3	(27)	
		MCDB	6.9	10.4	16.5	22.6	23.6	27.2	27.0	27.8	26.2	21.5	15.5	9.2	(28)	
	2%	WB	3.5	5.0	8.2	12.3	16.0	19.2	19.9	20.0	17.0	13.8	10.0	4.7	(29)	
		MCDB	4.6	7.4	13.2	19.3	22.2	25.3	25.9	27.0	23.9	18.5	12.9	5.9	(30)	
	5%	WB	2.0	3.6	6.7	10.9	15.1	18.2	19.0	19.0	15.9	12.5	8.3	3.1	(31)	
		MCDB	3.0	5.5	10.6	16.4	21.2	24.0	25.3	25.7	22.0	16.6	10.3	3.9	(32)	
	10%	WB	0.7	2.2	5.2	9.8	14.2	17.2	18.2	18.1	14.8	11.2	6.6	1.5	(33)	
		MCDB	1.4	3.5	8.1	14.4	19.8	22.5	24.3	24.5	20.2	14.4	8.2	2.1	(34)	
(35) Mean Daily Temperature Range	MDBR		8.7	10.4	11.4	13.3	14.0	13.8	14.6	15.4	14.6	13.7	9.5	7.3	(35)	
	5% DB	MCDBR	8.1	11.4	16.9	19.3	18.6	18.0	19.2	19.8	20.5	18.8	14.3	8.6	(36)	
		MCWBR	6.4	8.2	10.5	10.6	9.6	9.2	9.2	9.8	11.1	12.1	10.1	7.0	(37)	
	5% WB	MCDBR	7.4	10.9	14.8	16.8	16.6	16.2	16.4	17.7	17.8	17.3	13.0	8.0	(38)	
		MCWBR	6.0	8.0	9.6	9.5	8.9	8.6	8.6	9.2	10.0	11.3	9.4	6.7	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.311	0.344	0.393	0.458	0.430	0.425	0.447	0.459	0.410	0.369	0.340	0.312	(40)	
	taud		2.276	2.194	2.136	2.008	2.164	2.221	2.166	2.124	2.222	2.325	2.340	2.315	(41)	
	Ebn at Noon		768	810	820	796	838	843	817	788	795	776	728	723	(42)	
	Edh at Noon		80	106	131	163	145	137	143	144	119	93	76	69	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.30	2.13	3.26	4.36	5.44	5.84	5.84	5.34	3.81	2.57	1.44	1.01	(44)	
	RadStd		0.17	0.24	0.36	0.43	0.49	0.50	0.49	0.45	0.41	0.31	0.17	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.68	N/A	N/A	+0.93	+0.49	N/A	-210	-291	+112	+19	(46)
(47)	Regional (64 neighbors)	+0.66	N/A	N/A	+0.78	N/A	N/A	-146	-200	+78	+19	(47)

Nomenclature: See separate page